P510/1
PHYSICS
Paper 1
July - August 2026
3 hours

KAMSSA JOINT MOCK EXAMINATIONS

Uganda Advanced Certificate of Education

PHYSICS

Paper 1

3 hours

Instructions to candidates

- *This examination paper consists of **two** sections; **A** and **B**.*
- *It has **six** items. Section **A** has **two** compulsory items.*
- *Section **B** has **two** parts; **I** and **II**. Respond to only **one** item from each part.*
- *Respond to **four** items in all.*
- *Any additional item(s) responded to, will not be scored.*
- *Responses to each item should start on a fresh page.*

Silent non-programmable scientific calculators may be used.

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SECTION A *(compulsory)*

ITEM 1

In many display systems, the screen brightness should match the surrounding light conditions so that the picture remains clear and comfortable to watch.

A school has an older television set that uses a Cathode Ray Tube (CRT) and the picture is produced when electrons strike a fluorescent screen. The screen gives normal brightness if the electrons strike the screen with a velocity of $5.2 \times 10^{7} ms^{-1}$ in normal bright day light. The brightness depends on the energy and number of electrons reaching the screen.

During one of the digital electronics lessons, the teacher of S6 physics class made his students to watch the old Black and white CRT television in the school laboratory. The TV had been modified with an automatic brightness control system using a photosensitive plate fixed on top of the TV. The resulting photoelectric current is sent to a control circuit that controls the gride voltage of the CRT.

During the session, students noticed that when the windows are all open (bright day light), the screen becomes bright and when the windows are closed and laboratory curtains drawn, the screen became dimmer. The students did not understand how the CRT in the television produces a visible image and how the screen brightness changes. The room lighting is later changed to orange-red light that was very bright but the brightness on the screen remained unchanged. The students became even more confused about the situation.

Amina a student in the same class wonders whether the idea of moving electrons is used in any other devices.

Support material.

- Accelerating voltage of the CRT = $7.70KV$
- Work function of the photosensitive plate= $1.43eV$
- Planck's constant $h = 6.63 \times 10^{-34} Js$
- Charge of an electron $e = 1.6 \times 10^{-19} C$
- Mass of an electron $m = 9.11 \times 10^{-31} kg$
- Frequency of orange-red light = 2.9×10^{14}Hz

Task:

(a) As a physics student, analyze the information and determine if the screen was originally producing its normal brightness, and give response to students 'worries.

(b) Provide a response to Amina with clear explanation.

ITEM 2

During many sports competitions, officiation errors usually cause arguments and most times cause fights. In recent times, there has been introduction of video technology to help in minimizing human officiation errors.

In the final moments of a basketball game, a player begins to sprint from one end of the court when 3.50s remain to the end of the game clock. The player runs in a straight line, starting from rest and accelerating uniformly towards the center line, which is 13.7m horizontally from the hoop and immediately projects the ball on reaching that point. The ball enters the hoop after the buzzer (end of game signal/ whistle). The referee was not sure if the basket/ score should count, and decided to consult video replay.

A replay video captured the instant the player started sprinting from end of the court and the instant the ball entered the hoop but the instant when the ball left the players hands was not visible. However, the replay shows that the ball entered the hoop 5.0 s after the sprint began.

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The ball was released from a height of 2.4m and passed through a hoop at a height of 305cm. At the closing ceremony, David a former professional basketball player advised the players to concentrate at perfection of jump shots (where a player jumps up and holds a stationary position before releasing the ball), perfect smooth follow through (where a player throws the ball by extending arms pointing towards the target/hoop) and snap shots (where the fingers should create back spin on the basketball as its being released).

Support material:

- The basketball was released at an angle of 45° to the horizontal with a velocity of $11.5ms^{-1}$.
- The basket counts only if the ball is released before the buzzer.
- Air resistance is negligible.

Task: As a physics student, use the principles of mechanics to:

(a) Determine whether the score is valid, analyze and explain the motion of both the player and the ball.

(b) Explain the closing remarks of David.

SECTION B.

PART I

Respond to one item from this part

ITEM 3

At a renewable energy research station near Lake Victoria, intern students are assisting Engineers to investigate whether lake waves can support a small-scale Wave power project. The engineers explain that the available wave power device can only generate electricity if the wave amplitude is at least 0.25m. the students therefore need to determine whether the lake waves are suitable for the project.

The lake wave is modelled by the equation, $y = A\sin(2\pi ft - kx)$, where f is the frequency of the wave and k is the wave number, $y(m)$ is displacement of the water waves from undisturbed surface, and $x(m)$ is the distance moved by the water waves. For future research purposes, the students would like to have a record of the wave equation of the water waves.

Because the direct measurement of the moving waves is difficult, the students decided to use a screen method based on reflection of light to determine amplitude. A narrow laser beam is directed on to the water surface; the reflected beam falls on a vertical screen placed 5.0m away. When light is reflected from the crest of the wave (inclination of reflecting surface is zero), it reaches mean position O on the screen. When light is reflected from the wave surface at its greatest positive slope (inclination of the reflecting surface is β), the reflected spot shifts 0.8m above the mean position O. The students need to use knowledge of light reflection on rotated reflecting surface to determine the inclination of greatest slope β of the wave.

To check the accuracy of the students' results, Engineers use a second method. A converging lens is placed in the path of the reflected beam so that the light is focused into an optical Fiber. The optical Fiber carries the light to a photodetector and the electronically recorded signal is compared with the students' screen measurements. The students do not know how the optical Fiber transmits the light and why a convex lens was used in this option by the Engineers.

Support material.

- The angle of inclination of the greatest slope of the wave is equal to the maximum slope of the wave $\left(\frac{dy}{dx}\right)\max = |AK| = \beta$

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- Wavelength $\lambda = 20.0m$, *and period* $T = 5.0s$.
- Assume small angle conditions in radians apply.

Task: As a physics student:

(a) Analyze the information and help the students to find out if the project would be supported by the lake waves, and determine the lake wave equation.

(b) Discuss the accuracy check method used by the Engineers to address students' concerns.

ITEM 4

Uganda, just like any other African country, is facing the challenge of deforestation due to many households depending on the use of firewood and charcoal. Several projects and initiatives across Africa utilize concave/parabolic reflectors for solar cooking. In a bid to help nearby villages that are off-grid, a group of students were tasked with a project of designing a system that can help the villages to cook using concave/parabolic reflectors in which they needed to know the properties of the radiation.

The following information was provided:

- Stefan's Constant, $\delta = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}.
- Heat Capacity of the thermometer bulb = 70J K^{-1}.
- Effective cooking system should take not more than 105minutes.

One group designed a system which employs a digital thermometer with a blackened spherical bulb of diameter 10 mm, initially at 20°C that has to be placed at the focus of a concave reflector of diameter 0.2m, and did not know how to locate it. The reflector is directed towards the Sun and the Solar Power incident on the reflector is 100Wm^{-2}. The group members were not sure whether this system will be effective, the practical improvements for improving its efficiency, and they also needed to understand if there's any another aspect in which this renewable energy is used apart from the use of concave reflector.

Task:

a) Evaluate the effectiveness of the system designed by the group.

b) Propose, with justification, a real-life application of another aspect the group members were interested in.

PART II

Respond to one item from this part

ITEM 5

Your class was taken to one of the battery companies in town. Of late, customers are complaining about the quality of the batteries produced. This prompted the company manager to design an investigation to verify customers' complaints using two quality checks.

In the first check, a battery is connected in series with two resistors of resistances 15Ω and 27Ω. The potential difference across the 15Ω resistor was balanced against the potential drop across 779 mm of a uniform potential wire carrying a steady current. The 27Ω resistor was then replaced with 35Ω resistor while keeping the 15Ω resistor unchanged. The length of the same potentiometer wire carrying the same amount of current required to balance the potential difference across the 15Ω resistor was found to be 656mm.

In the second check, an ammeter is connected to a 30Ω resistor and a setup which has a soft iron ring with two coils A and B. The battery is to be connected to an inverter which changes d.c voltage to a.c voltage of r.m.s value equal to the e.m.f of the battery Coil B has turns three

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times that of Coil A. Coil A is to be connected across the inverter while B is to be connected in series with the ammeter and the resistor. The ammeter reading of 0.79A is expected. The manager told the learners that he anticipates that the resistor will heat up after some time.

The manager further guided that for their type of battery to be pronounced to be good, it should have an internal resistance not exceeding 0.9Ω and its *e.m.f* should be in the range of 6.0 – 10.0 V.

The manager shared this information with the learners as a way of challenging them to see if they mastered the concepts they had already learnt in classroom.

Task:

a) Analyze the information provided to verify customers' complaints.
b) Interpret the operation and a real-life application of the device having Coils A and B

ITEM 6

A certain school has been experiencing a challenge of being supplied with fake school laboratory equipment. The school procurement committee has come up with a regulation of testing any equipment supplied to the school before it is accepted. A small Van de Graaff generator was supplied to the school; the generator uses a motor driven insulating belt to transfer charge to a spherical metal dome of radius 0.12m. The school laboratory technician did not know how the system works and even what to check for. According to the operating manual, charge of $2.4 \times 10^{-6}C$ can be obtained on the dome if the motor driving the generator belt operates at 12V and draws a current of 0.5A for one minute, and at least 0.05% of the electrical energy should be stored as electrostatic energy. The spherical dome acts as an isolated spherical capacitor.

Davis a senior six physics student was requested to work with the laboratory Technician. Davis suggested that they can test for charge distribution and how much of the motor electrical energy is converted to stored electrostatic energy. During the test, the generator is kept running for more than one minute. At some time, sparks were observed around the dome. This surprised both Davis and the technician. Before the two could complete the test, Davis was called for his next lesson.

Support material:

- Permittivity of free space $\varepsilon_O = 8.85 \times 10^{-12} Fm^{-1}$
- Dielectric strength of air $=3.0 \times 10^{6} Vm^{-1}$.

Task:

(a) As a physics student, analyze the information, and help the laboratory technician to understand the working of the system and make a decision on whether to accept the generator or not.
(b) Explain what could have caused the sparks around the dome, with clear details to convince Davis.

END

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